Assignment module 6: Network Security, Maintenance, and Troubleshooting Procedures

Section 1: Multiple Choice

1. What is the primary purpose of a firewall in a network security infrastructure?

   a) Encrypting network traffic
   b) Filtering and controlling network traffic
   c) Assigning IP addresses to devices
   d) Authenticating users for network access

   Ans: Filtering and controlling network traffic

2. What type of attack involves flooding a network with excessive traffic to disrupt normal operation?

   a) Denial of Service (DoS)
   b) Phishing
   c) Spoofing
   d) Man-in-the-Middle (MitM)

   Ans: Denial of Service(DoS)

3. Which encryption protocol is commonly used to secure wireless network communications?

   a) WEP (Wired Equivalent Privacy)
   b) WPA (Wi-Fi Protected Access)
   c) SSL/TLS (Secure Sockets Layer/Transport Layer Security)
   d) AES (Advanced Encryption Standard)

   Ans: SSL/TLS (Secure Sockets Layer/Transport Layer Security)

4. What is the purpose of a VPN (Virtual Private Network) in a network security context?

   Ans: Main purpose of VPN is:

   - Over the internet ,creates the tunnel
   - Secure, encrypted data transfer
   - Masking the IP address
   - Protecting online privacy

Section 2: True or false

True or False: Patch management is the process of regularly updating software and firmware to address security vulnerabilities and improve system performance.

Ans: True

True or False: A network administrator should perform regular backups of critical data to prevent data loss in the event of hardware failures, disasters, or security breaches.

Ans: True

True or False: Traceroute is a network diagnostic tool used to identify the route and measure the latency of data packets between a source and destination device.

Ans: True

Section 3: Short

8. Describe the steps involved in conducting a network vulnerability Assignment.

Ans: Steps involved are:

- Scanning for vulnerability and weakness,
  Detect security loopholes in network devices like router,switches,servers. misconfigurations , outdated software

- Risk Evaluation,
  Prioritize the most high vulnerabilities and addressing those pose the greatest risk to network security.

- Patch Management, Troubleshooting,
  Take action by reconfiguring the system or deploy security controls against vulnerabilities.

- Reporting ,record of action taken.
  A detail report of vulnerability found , type of risk , and step taken to rectify the network vulnerability must be record for future and for similar situations in future time.

Section 4: Practical Application

9. Demonstrate how to troubleshoot network connectivity issues using the ping command.

Ans:

```
:\Users\User>ping google.com

inging google.com [142.250.76.174] with 32 bytes of data:
eply from 142.250.76.174: bytes=32 time=19ms TTL=118
eply from 142.250.76.174: bytes=32 time=21ms TTL=118
eply from 142.250.76.174: bytes=32 time=20ms TTL=118
eply from 142.250.76.174: bytes=32 time=20ms TTL=118

ing statistics for 142.250.76.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
pproximate round trip times in milli-seconds:
    Minimum = 19ms, Maximum = 21ms, Average = 20ms

:\Users\User>
```

It is used to check the connectivity of the system in the network.

To check the latency in the network

Command used :- ping [IP address of destination]

10. Discuss the importance of regular network maintenance and the key tasks involved in maintaining network infrastructure.

Ans:

Importance of regular maintenance are:-

- Keeps network running smooth and with quick responsive.
- Protects system from threats and damage to equipments.
- Keeps ready for future addition of new devices to old infrastructure.

Keys task involved in maintaining network infrastructure are:-

- Monitoring performance
- Keep updated with required tools and patches
- Check security settings
- Check connectivity with ping command

Section 5:

1. Which of the following best describes the purpose of a VPN (Virtual Private Network)?
   a) Encrypting network traffic to prevent eavesdropping
   b) Connecting multiple LANs (Local Area Networks) over a wide area network (WAN)
   c) Authenticating users and controlling access to network resources
   d) Reducing latency and improving network performance

Ans: Encrypting network traffic to prevent eavesdropping